

Probabilistic Programming

Final Exam

Question 1 (0.5 points). Let us assume a probability distribution for the time between events in a process in which events occur continuously and independently at a constant average rate. Which of the following types of distribution would we choose for modelling:

- a) Binomial distribution
- ☒ b) Exponential distribution
- c) Poisson distribution

Question 2 (0.5 points). From the following distribution which one is a conjugate prior for the Poisson distribution (likelihood):

- ☒ a) Gamma distribution
- b) Beta distribution
- c) Normal distribution

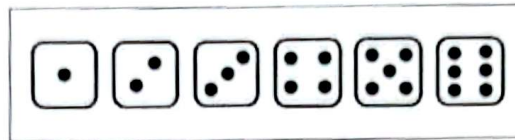
Problem 1 (1.5 points). Cards can be used instead of dice. Thus, the eleven outcomes of two six-sided dice: 2, 3, ..., 12 (the sum of the two dice) are paralleled by the outcomes of randomly drawing two cards from a deck of 12 cards: A-6 of hearts and A-6 of spades and summing them (A is representing 1). Are these two methods equivalent? That means, have the two methods produced the same distribution of the outcomes? Justify / prove your answer.

Problem 2 (1.5 points). The "Under and Over 7" is an old carny game often seen at fairs and Las Vegas nights, but rarely found in any casino. The Under and Over 7 game is a betting game played with two six-sided dice. The game's objective is to predict whether the sum of the two dice will be under 7, over 7, or exactly 7. The banker pays out based on the roll result and each players' bets:

- Under 7: If the total is under 7, the banker pays 1 for 1 on this bet. If not, the player loses their chips.
- 7: If the total is exactly 7, the banker pays a higher payout of 2 for 1. Otherwise, the player loses.
- Over 7: If the total is over 7, the banker pays 1 for 1. If not, the player loses.

If player bets 1\$, write a single PyMC stochastic variable that can represent the stochastic output of the game: 0 (losing), 1 (winning 1\$), 2 (winning 2\$).

Problem 3 (2 points). MY AUNT - YOUR AUNT is an infamous traditional gambling game in Germany, there called MEINE TANTE – DEINE TANTE. The game is usually played with cards, but the dice version has its followers. You need two dice and the following playing board:

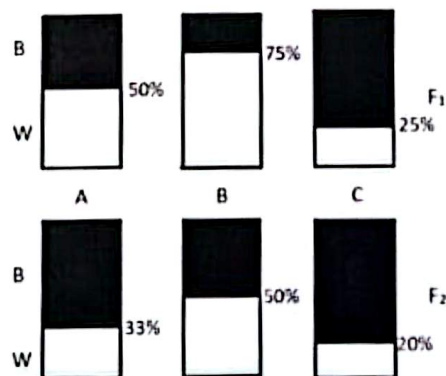


The aim of the game is to win against the banker by successfully betting on numbers rolled for 'Your Aunt' and avoiding numbers rolled for 'My Aunt'. First, you place your bet on one number on the board. It may take several turns until your bet is decided. The banker throws one die to left commenting 'My Aunt' and one die to the right 'Your Aunt':

- If 'My Aunt' shows the number of your bet, you lose.
- If 'Your Aunt' shows the number of your bet and 'My Aunt' does not, you win.
- If both 'My Aunt' and 'Your Aunt' show the number of your bet, you lose.
- If your bet does not occur on any of the dice, your bet remains on the board for the next throw until the game is decided.

Write a (complete) PyMC program that estimates the probability of winning of a specified bet (1, 2, 3, 4, 5, 6).

Problem 4 (3 points). Alice has 3 cards A, B, and C each with two faces: F1 and F2. Each card has on each face a mixture of black and white coloring in a given ratio, as can be seen in the figure below:



Bob picks a card at random, places it randomly with a face on the table, and points at a random location on the card with a pen. Bob tells Alice the pen landed on white. Help Alice to write the full (complete) PyMC program that estimates the (posterior) probability distribution of the probability that Bob picked one of the 3 cards A, B, C.